

PORI, Finland

WMO: 029520

Lat: 61.467N

Lon: 21.788E

Elev: 44

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.9	-0.3	-10.6	3.1	-6.0	-4.4	4.4	0.5	23.5	33.5	20.9	32.8	4.0	130	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	80.3	64.9	76.8	63.3	73.3	61.6	67.3	75.8	65.3	73.2	63.4	71.0	8.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	90.7	70.6	62.3	84.1	69.2	60.4	78.5	67.0	31.8	75.4	30.1	72.9	28.8	70.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	17.9	15.9	DB	-9.9	84.9	7.5	4.3	-15.3	88.0	-19.7	90.5	-23.9	93.0	-29.4	96.1	(4)
(5)			WB	-10.3	70.1	7.4	2.6	-15.6	72.0	-20.0	73.5	-24.1	75.0	-29.5	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	23.5	23.0	29.0	39.2	49.0	56.7	62.7	60.4	52.0	41.8	34.2	28.3
	DBStd	15.78	10.83	11.06	7.83	6.28	7.23	5.90	5.42	5.31	5.88	7.10	7.89	10.24
	HDD50	4132	820	755	652	330	106	8	0	1	47	265	475	673
	HDD65	8554	1285	1175	1117	774	499	257	111	162	391	720	925	1138
	CDD50	1123	0	0	0	6	74	210	394	323	106	10	0	0
	CDD65	70	0	0	0	0	2	9	40	19	0	0	0	0
	CDH74	625	0	0	0	0	33	98	342	151	1	0	0	0
	CDH80	106	0	0	0	0	3	14	71	18	0	0	0	0
Wind	WSAvg	7.3	7.6	7.5	7.6	7.3	7.6	7.6	6.8	6.5	6.8	7.0	7.6	8.0
Precipitation	PrecAvg	24.1	2.0	1.4	1.3	1.1	1.3	2.2	2.6	2.9	2.2	2.8	2.2	2.1
	PrecMax	29.7	3.3	3.1	2.3	2.7	3.4	4.4	4.9	6.1	4.8	6.5	5.6	4.1
	PrecMin	20.3	0.3	0.0	0.2	0.0	0.2	0.7	0.1	1.2	0.4	0.8	0.4	0.2
	PrecStd	2.4	0.8	0.8	0.5	0.6	0.8	1.0	1.3	1.3	1.4	1.2	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	42.4	50.1	66.6	78.6	82.1	86.2	82.4	71.3	59.2	51.5	46.6
		MCWB	39.9	38.7	41.5	51.6	60.2	64.3	68.0	66.9	60.6	54.3	49.3	45.1
	2%	DB	39.4	39.5	44.6	59.2	71.9	76.7	81.4	78.2	67.6	55.7	48.3	44.4
		MCWB	37.8	37.3	39.2	48.0	56.5	61.8	66.0	64.4	59.1	52.8	46.8	43.1
	5%	DB	37.6	37.6	41.2	55.0	66.9	72.0	77.4	74.7	64.4	53.8	46.4	42.4
		MCWB	36.3	36.0	37.1	45.5	54.6	59.0	64.1	63.1	57.5	51.2	45.3	41.1
	10%	DB	35.8	35.7	38.9	50.7	62.4	68.3	73.8	71.4	61.7	51.9	44.5	39.6
		MCWB	34.8	34.4	35.7	43.3	52.3	57.4	62.6	61.9	56.6	49.5	43.5	38.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	39.3	42.7	53.0	62.9	67.2	70.7	69.7	62.3	55.5	49.8	45.4
		MCDB	41.2	40.5	48.3	63.0	73.6	77.9	81.3	77.0	68.2	58.2	50.6	46.1
	2%	WB	37.9	37.5	39.6	49.2	59.4	63.8	68.1	66.8	60.4	53.7	47.0	42.9
		MCDB	38.7	38.9	43.2	57.7	69.1	72.9	77.1	74.3	65.5	55.7	47.9	43.5
	5%	WB	36.3	36.0	37.6	46.5	55.8	61.1	66.0	64.7	58.8	52.0	45.1	40.5
		MCDB	37.2	37.3	40.4	53.0	64.6	69.1	74.0	71.9	63.2	54.0	45.8	41.4
	10%	WB	34.8	34.4	35.9	43.8	52.9	58.9	64.1	63.0	57.2	50.1	43.2	38.5
		MCDB	35.5	35.4	38.1	49.6	60.7	66.2	72.2	69.4	61.0	51.5	44.0	39.4
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.7	13.0	15.8	17.6	16.2	16.4	16.6	14.6	10.5	7.6	8.7
		MCDBR	8.3	9.5	14.8	23.7	24.5	23.3	22.9	22.2	18.2	11.8	7.6	7.8
		MCWBR	7.7	8.2	10.3	13.7	12.3	11.3	10.0	10.7	10.8	9.2	7.0	7.3
	5% WB	MCDBR	7.8	8.8	12.5	21.3	21.1	19.7	19.2	18.5	15.1	10.5	7.4	7.3
		MCWBR	7.5	7.8	9.4	13.1	11.6	10.8	9.7	10.1	9.8	8.7	7.0	6.9
Clear-Sky Solar Irradiance	taub	0.264	0.309	0.329	0.368	0.365	0.357	0.383	0.370	0.345	0.327	0.284	0.352	
	taud	2.136	2.235	2.317	2.327	2.410	2.463	2.413	2.440	2.496	2.395	2.191	2.530	
	Ebn at Noon	140	208	249	261	272	276	264	259	245	203	130	91	
	Edh at Noon	12	21	27	33	32	31	32	29	23	18	11	8	
All-Sky Solar Radiation	RadAvg	62	239	721	1226	1715	1870	1728	1348	800	331	85	36	
	RadStd	9	49	84	128	151	118	155	126	87	57	16	4	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (50 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page